INCH-POUND

MIL-C-2181M
27 September 1990
SUPERSEDING
MIL-C-2181L
14 December 1987

MILITARY SPECIFICATION

COVERALLS, TOXICOLOGICAL AGENTS PROTECTIVE, M-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This document covers a one piece coverall for the protection of personnel engaged in extremely hazardous decontamination work or other special operations involving danger from spillage or splashing of liquid chemical agents. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order (see 6.2).

1.2 Classification. The coverall shall be of one type in the following sizes, as specified (see 6.2).

Schedule of sizes

Small
Medium
Large
X-Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-T-285	-	Thread, Polyester
JJ-W-155	-	Webbing, Textile, (Cotton, Elastic)
QQ-B-613	-	Brass, Leaded, and Non-Leaded: Flat Products (Plate, Bar, Sheet, and Strip)
TT-E-529	-	Enamel, Alkyd, Semigloss, Low VOC Content
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polyropylene Foam
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-B-286	-	Buttons, Tack; and Tack, Button
MIL-B-371	-	Braid, Textile, Tubular
MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-C-3133	-	Cellular Elastomeric Materials, Molded or Fabricated Parts
MIL-L-3851	-	Loops, Slide (For Equipage)
MIL-S-6790	-	Suspenders, Trousers (Suspenders Flying Suit)
MIL-F-10884	-	Fasteners, Snap
MIL-C-12189	-	Cloth, Coated: Butyl Coated, Toxicological Agents Protective
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-406 - Plastics: Methods of Testing
FED-STD-601 - Rubber: Sampling and Testing
FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-670 - Classification System and Tests for Cellular Elastomeric Materials
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

Department of the Army Technical Manuals

TM 10-8400-203 - General Repair Procedure for Individual Clothing

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-204 - Standard Methods for Testing Sewing Threads
D-3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

* 3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Coated cloth. The coated cloth shall be cloth, nylon, butyl-coated, conforming to MIL-C-12189.

* 3.3.2 Strapping. The strapping used to seal the seams and hems shall be 1 ± 1/16 inch wide, true bias cut (45 degrees ± 10 degrees) and shall be made of the coated fabric specified in 3.3.1 or butyl coated cotton cloth conforming to MIL-C-12189.

3.3.3 Cement. The cement used to pre-cement the seams and to adhere the strapping to the seams shall be of the self-curing neoprene or butyl-base type. The cement, when applied and tested as specified in table IV and 4.4.1.1, shall not crack or flake and the hydrostatic resistance of the cemented fabric shall be equal to that of the butyl-coated cloth before cementing.

3.3.4 Sponge rubber. Sponge rubber for the sleeve pad shall be 1/4 ± 1/32 thick and shall conform to MIL-C-3133 and to type R, grade R03 of MIL-STD-670.

3.3.5 Thread.

* 3.3.5.1 For seaming and stitching. The thread shall be polyester, type I, class 1, subclass A conforming to V-T-285. Sizes shall be B or E as specified in table II. The color of any seam or stitching not covered by strapping shall be Olive Drab S-1, C.A. 66022 and colorfastness is not required.

3.3.6 Webbing elastic. The elastic webbing shall be type I, class 3, 3/4 ± 1/32 inch wide, unbleached, conforming to JJ-W-155. If the above referenced is not available a commercial elastic webbing 3/4 inch in width weighing not less

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than 0.40 ounces per linear yard may be substituted provided the initial tension is no less than 40 percent or more than 60 percent when tested as specified in JJ-W-155.

3.3.7 Webbing cotton. The cotton webbing shall be type IIa, class 4, 3/4, 1 or 1-1/2 inches wide as applicable, dyed Olive Drab 7 conforming to MIL-W-530.

3.3.8 Braid for gusset. The braid for the gusset shall be type V, class 4, Olive Drab 7 conforming to MIL-B-371. It shall be 85 inches long and tipped or resin coated at both ends, to prevent raveling.

3.3.9 Buttons, metal. The metal buttons shall be type I or II, size 27 line and the tack shall be type IX, size 14 or 17 line by 0.406 conforming to MIL-B-286.

3.3.10 Suspenders. The suspenders shall be type I of MIL-S-6790 except that the wire buttonhole of the suspenders shall be a tight fit on the shank of the selected metal button specified in 3.3.9, the color of the webbing shall be Olive Drab 7.

3.3.11 Buckle. The buckle shall conform to type II, style 4, class 1, construction B, size 3/4 inch of MIL-B-543, except the brass shall conform to composition 268, quarter-hard, or half-hard brass of QQ-B-613.

* 3.3.12 Snap fasteners. The snap fasteners shall be fastener, snap male and female conforming to style 2A for suspenders only and style 2 for all others, finish 2 of MIL-F-10884. The size of the button and eyelet barrels selected shall allow a secure clinch of the snap after the components are assembled.

* 3.3.13 Eyelet and washer. The eyelet and washer shall conform to Stimpson eyelet number A798, or equivalent.

3.3.14 Loop slide. The loop slide shall be steel and conform to class 3, style 2, construction A, size 2 inch of MIL-L-3851.

3.3.15 Clasp. The clasp assembly shall be steel conforming to North and Judd No. 1375 or equal. The finish shall consist of electroplated zinc followed by a phosphate surface treatment and baked enameling. The enamel finish shall conform to type II of TT-E-529 and shall be black.

3.3.16 Slide keeper. The slide keeper shall measure 1-1/4 $\pm$ 1/16 by 3/4 $\pm$ 1/16 by 1/8 $\pm$ 1/32 inches and shall be fabricated from a vinyl polymer or copolymer, with an "A" scale Shore durometer hardness of 70 $\pm$ 10. Two 1/4-inch round holes shall be located in the centerline parallel to the 3/4 inch dimension each hole being centered 5/16 inch from each end of the keeper. Rounded corners are permitted.

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3.3.17 Identification label. The identification label shall be either a cloth label or a stamped label containing the following information:

Nomenclature
Size
Name of contractor
Date of manufacture and number of contract
Specification number

3.3.17.1 Cloth label. When a cloth label is used, it shall conform to type VI of DDD-L-20, with the following exceptions:

a. The label shall be coated as specified for classes 3, 7 and 11.
b. The label shall measure 2-3/4 by 1-1/2 inches and the legend shall be printed parallel to the small dimension.
c. All colorfastness and writing tests are not applicable.

3.3.17.2 Stamped label. When a stamped label is used, the legend shall be printed directly on the coated fabric with black indelible ink. Letters shall be sans-serif, extra bold, monotone style, 14 point (caps only).

3.3.17.3 Colorfastness of labels. Both the cloth label and the stamped label shall be legible and shall not become faded or tacky after testing as specified in 4.4.1.1.

3.3.18 Dusting powder. The dusting powder for application to the coveralls shall be whiting (less than 1.0 percent free silica) talc (nonfibrous), or other finely divided low toxicity material which does not support mildew growth.

* 3.3.19 Soap. The soap for seam testing shall be an anionic surfactant 2.5 weight percent (e.g.; IGEPON-T77 or equivalent) to be used in distilled water (see 6.5).

3.4 Patterns. Standard patterns which provide the proper allowance for all seams, will be furnished by the Government. The Government patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns.

3.4.1 Pattern parts. The pattern components for the coveralls shall be cut from the materials as specified and according to the number of parts listed in table I.

TABLE I. List of pattern parts

Item	Nomenclature of part	Cut parts
Butyl coated cloth	Body	2
	Right sleeve	1
	Left sleeve	1

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TABLE I. List of pattern parts (cont'd)

Item	Nomenclature of part	Cut parts
Butyl coated cloth (cont'd)	Right oversleeve	1
	Left oversleeve	1
	Outside waist band (right and left)	2
	Inside waist band (right and left)	2
	Gusset	1
	Gusset facing	1
	Belt	1
	Belt loop holders	6
	Outside crotch reinforcement patch	1
	Collar	1
	Neck drawcord loop	1
	Sleeve and oversleeve adjustment straps	4
	Suspender button assembly cover	6
	Leg adjustment strap	2
	Right leg strap assembly patch	1
	Left leg strap assembly patch	1
	Oversleeve stud reinforcement cover	2
	Sleeve and leg stud reinforcement cover	4
Sponge rubber	Sleeve pads	2

NOTE: Pattern parts are not furnished for the suspender reinforcement piece, stud reinforcement piece for leg and sleeve, stud and cap reinforcement pieces, and elastic webbing for leg strap (see table II, operation 1).

3.5 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figures 1 and 2 are furnished solely for guidance and information. Should variation from the document appear in figures 1 or 2, the document shall govern.

3.5.1 Stitching, seams and stitches. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the coveralls.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

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3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner.

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1 a. and b. Type 301 stitching may be used to perform the repairs.

3.5.1.3 Bartacks. Unless otherwise specified, all bartacks shall be 3/8 ± 1/16 inch long and 1/8 ± 1/32 inch wide, and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose stitching.

* 3.5.1.4 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in Table II, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government finished shaper pattern is forwarded, the component shall conform to that pattern.

3.5.1.5 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.5.2 Pre-cementing. All joining seams shall be overlapped 3/8 ± 1/16 inch and cemented before stitching.

3.5.3 Seam strapping.

3.5.3.1 Preparation of seam area for strapping. Prior to application of the cement, the seam, the area along the seam where the strapping is to make contact, and the surface of the strapping to be adhered, shall be thoroughly cleaned with rubber makers naptha or other suitable cleaning agent which is not a solvent for the butyl coating compound. In addition, the above procedure shall be employed for all cementing and pre-cementing operations.

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* 3.5.3.2 Application of seam strapping. Strapping shall be applied to both the inside and outside of all seams, using the cement specified in 3.3.3. Each individual seam shall be strapped with a continuous (single piece, non-overlapping, seam length) length of strapping. Where ends of strapping meet, a 1 ± 1/4 inch overlap shall be made. Wherever strapping intersect, they shall be overlapped. Butting or piecing shall not be permitted. When the strapping is centered over the seam and pressed into position, the cement line may extend a maximum of 1/2 inch on either side of the strapping.

3.5.4 Setting of snap fasteners. A hole shall be prepunched through the materials to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.5.5 Setting of eyelets. A hole shall be prepunched before inserting the eyelet. The hole shall be smaller than the outside diameter of the eyelet barrel so that the barrel must be forced through the hole. The eyelet shall be securely clinched without splitting in a manner that will prevent detachment from or cutting of the surrounding material. Unless otherwise specified the washer may be on either side of components on which installed.

3.5.6 Repairs in coating on finished coverall. Areas containing coating defects may be repaired by application of two round, superimposed, 1-1/2 ± 1/4 inch diameter patches with one patch on the inside of the coverall and the other on the outside of the coverall. The patches shall be made of the coated cloth specified in 3.3.1 and shall be adhered with the cement specified in 3.3.3. The lightly coated side of the patch shall be cemented to the coverall and the patch shall extend a minimum of 1/2 inch beyond all edges of the damaged area. The patch shall be smoothly and evenly adhered over its entire area, and shall be well adhered to the coated cloth so that, when subjected to flexing action between the hands, will show no lifting of the edge, and no cracking, flaking or removal of the cement or patch. The repaired area shall show no stiffening or tackiness which would affect the serviceability. When patching areas where there is a double layer of the coated cloth, e.g., front facing, only one repair patch is needed to cover the damage. It is not necessary to place a patch on the reverse side of the damaged area.

3.5.6.1 Location and number of repairs. Repairs not exceeding eight in number may be made to each coverall, but are limited to the following components and amounts; not more than two on either sleeve, nor more than three on either body pattern part (left or right), or gusset.

3.6 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with all operations specified in table II. The contractor is not required to follow the exact sequence of operations. The use of drill holes is prohibited except for marking locations of snap fasteners and eyelets and shall be no larger than 1/16 inch in diameter.

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1 On joining seams, score as open seam

TABLE II.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
*	a. The coverall shall be cut in strict accordance with the patterns furnished which show size and notches for proper assembly of all parts and placement of snaps and buttons. All component parts of the coverall shall be matched to ensure a uniform shade throughout the garment. All component parts of the coverall shall be cut and assembled so that the heavily coated side shall be on the outside of the coverall. The strapping shall be 1 ± 1/16 inch bias cut as required for seams. No pin tickets shall be used.						
	b. Cut six pieces of 1-1/2 inch webbing 1-1/2 ± 1/16 inches in length, to be used for suspender button reinforcement pieces.						
	c. Cut four pieces of 1 inch webbing 3-1/4 ± 1/16 inches in length, to be used for sleeve and leg stud reinforcement pieces.						
	d. Cut two pieces of 1 inch webbing 2-3/4 ± 1/16 inches in length, to be used for the oversleeve stud reinforcement piece.						
	e. Cut two pieces of 3/4 inch wide elastic webbing 6 ± 1/16 inches in length, to be used to make the leg adjustment strap.						
2.	Shade and size marking.						
	Mark, ticket, or bundle all components, except those cut from ends, to ensure that the heavily coated side of the fabric is on the outside of						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Shade and size marking. (cont'd)						
	the coveralls and that uniform shade and size is maintained throughout the coveralls.						
3.	Replacement of defective or damaged parts.						
	During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.4 shall be removed from production and replaced with non-defective and properly matched components.						
4.	Pre-cementing.						
	All joining seams shall be pre-cemented (see 3.5.2).						
5.	Make sleeves.						
	a. Make one adjustment strap for each sleeve and one for each oversleeve as follows:						
	(1) Fold piece in half as indicated by fold line on pattern and sew together around all sides.	301	SSa-1	7-9	B	B	
	(2) Attach one snap cap to one end as indicated on pattern.						
	(3) Pre-cement both sides of the end of the adjustment straps without the snap cap. Cement strip shall be 3/4 ± 1/4 inch wide.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make sleeves. (cont'd)						
	(4) Attach adjustment straps to sleeves and oversleeves as indicated by notches on patterns 1/16 to 1/8 inch from edge. The	301	SSa-1	7-9	E	E	
	b. Turn inside and sew 1/2 inch wide hem on each sleeve and each oversleeve.	301 or 401	EFa-1	7-9	E	E	
	c. Stitch oversleeve to sleeve as indicated by marks on pattern.	301 or 401	LSbj-1	7-9	E	E	
	d. Seam inseam of sleeve and oversleeve with a single row of stitching, catching the sleeve straps in the seam. The tabs shall point toward the back of the coverall.	301 or 401	LSa-1	7-9	E	E	
	e. Strap all sleeve and oversleeve seams and hems on both sides with 1 inch strapping.						
	f. Superimpose sleeve stud reinforcement cover and sleeve stud reinforcement piece (1 inch wide webbing). Attach stud portion of snap fasteners at position shown on sleeve stud reinforcement cover.						
	g. Superimpose oversleeve reinforcement cover and oversleeve stud reinforcement piece (1 inch wide webbing). Attach stud portion of snap fastener at position shown on oversleeve stud reinforcement cover.						
	h. Cement stud assemblies referenced in "f" and "g" above to the sleeves and oversleeves at positions indicated on pattern.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make sleeves. (cont'd)						
	i. Cement sleeve pads at the bottom of each sleeve on the inside. The pad is not long enough to completely encircle the inner part of the sleeve and the area to be left open shall be where the strapped seam is located.						
6.	Make belt.						
	a. Cement entire width of belt and fold as indicated by notches on pattern.						
	b. Thread belt 1 to 1-1/2 inches over center bar of adjusting loop slide, with the seam on the underside of belt.						
*	c. Box stitch end of belt 1/4 to 1/2 by 1-7/8 ± 1/8 inches wide with ends securely tacked.	301	Box	7-9	E	E	
	d. Thread belt through female portion of the clasp with the hook on the outside of the belt. Thread free end of belt over center bar of adjusting loop slide. Finished belt shall fasten with hook facing away from body.						
*	e. Thread belt through male end of the clasp, turn under 1 ± 1/8 inch and box stitch end of belt 1/4 to 1/2 by 1-7/8 ± 1/8 inches wide, with the ends securely tacked.	301	Box	7-9	E	E	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make belt loop.						
	a. Fold the piece of material, overlapping the edges, and stitch through all pliles with one row of stitches.	301	EFy-1	7-9	B	B	
	b. Place one row of stitches 1/16 to 1/8 inch from each folded edge. The finished belt loop shall measure 3/8 to 1/2 inch wide.	301	EFy-2	7-9	B	B	
	c. Operation "b" above may be on a two needle machine in one operation.						
8.	Make and attach leg adjustment patch and strap.						
	a. Fold leg strap in half and position elastic webbing.						
	b. Seam one end of leg strap, catching the elastic webbing in the seaming. Continue stitching down the side of the leg strap allowing the elastic webbing to remain free.	301	SSa-1	7-9	B	B	
	c. Turn leg strap right side out.						
	d. Attach female cap to leg strap (through cover and elastic) as indicated by marks on pattern.						
	e. Draw free end of the elastic even with the tunnel edge (forcing the wrinkles in the leg strap cover toward the snap) and stitch across 1/8 inch from end through all plies.	301	SSa-1	7-9	E	E	
	f. Insert leg adjustment strap into the slot of the leg adjustment patch and box stitch (5/8 ± 1/16 inch) square in the center of patch.	301	Box	7-9	E	E	

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TABLE II. (cont')

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make and attach leg adjustment patch and strap. (cont'd)						
	g. Cement the assembly of patch and strap to the leg as shown on pattern.						
9.	Make gusset.						
	a. Stitch the gusset reinforcement piece to the under-side of the gusset as shown on pattern. Stitching shall be 1/16 to 1/8 inch from edge of facing.	301 or 401	SSAE-3	7-9	B	B	
	b. Attach 22 eyelets and washers to gusset as shown by marks on pattern.						
10.	Make neck drawcord tunnel.						
	a. Hem both ends of neck drawcord by turning 1/4 inch and stitching.	301	EFa-1	7-9	B	B	
	b. Fold neck drawcord loop in half widthwise and join by stitching along the open edge 1/8 ± 1/32 inch from edge.	301	SSa-1	7-9	B	B	
11.	Hem legs.						
	a. Turn inside and sew 5/8 ± 1/16 inch wide hem on legs.	301 or 401	EFa-1	7-9	E	E	
	b. Strap leg hems on both sides with 1 inch strapping.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Close inseam and attach strapping.						
	a. Seam inseam by bringing the front over the back and stitching in place.	301 or 401	LSa-1	7-9	E	E	
	b. Fully strap inside and outside of inseams. All bottom ends of outside strapping (i.e.,) sleeves and legs shall be returned to the inside of the hem not less than 1 inch.						
13.	Join backs or body.						
*	a. Begin at front and join right body to left body up through the back to the neck.	301 or 401	LSa-1	7-9	E	E	
	b. Strap body joining seam on inside and outside with 1 inch strapping.						
	c. Cement crotch reinforcement patch at junction of crotch seams and inseams with patch centered over junction area.						
	d. Superimpose leg stud reinforcement cover and leg Stud reinforcement piece (1 inch wide webbing). Attach stud portion of snap fasteners at position shown on leg stud reinforcement cover.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Join backs or body. (cont'd)						
	e. Cement stud assembly to legs at position indicated on pattern.						
14.	Make suspender button attachment, join attachments to inner waistband, and cement inner waistband to the body.						
	a. Fold suspender button assembly cover in half and place the suspender button reinforcement 1-1/2 inch webbing) between the two layers.						
	b. Attach metal button through the three pieces at center point of assembly.						
	c. Cut slots for buttons on inside waistband as indicated by marks on pattern.						
	d. Insert button through slit in inner waistband and stitch the assembly on all sides to the inner waistband with one row of stitching.	301	SSa-1	7-9	B	B	
	e. Repeat operation for all six suspender buttons.						
	f. Cement the inside waistband to the coverall in positions as shown by marks on pattern.						
15.	Attach waistbands and belt loops.						
	a. Cement outside waistband as indicated by mark on pattern.						
	b. Cut belt loops 4-1/4 ± 1/16 inches long.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Attach waistbands and belt loops. (cont'd)						
	c. Insert belt loops through slits in belt loop holder and tack through loop and patch 1/2 inch from each end or each slit.	301	SSa-1	7-9	B	B	
	-or-						
	A bartack may be used in lieu of a tack.	Bartack		28 per bartack	B or E	B or E	
	d. Cement the belt loop assembly patch as indicated by marks on pattern.						
16.	Attach sleeves.						
	a. Join sleeve to armhole properly distributing fullness.	301 or 401	LSa-1	7-9	E	E	
	b. Strap inside and outside of armhole with 1 inch wide bias strapping.						
17.	Attach gusset, facing, and reinforcement.						
	a. Cement a 1-1/2 inch wide strip of the same coated cloth as the coveralls (see 3.3.1) on either one side or the other of the fold line on the front facing and to the inside of both the right and left fronts. It shall be placed 1/8 to 3/8 inch from the fold line, 3/8 to 5/8 inch from both top and bottom of the facing and centered over the snap fastener punch holes.						
	NOTE: The facing should be centered on both sides of the reinforcement strip.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Attach gusset, facing, and reinforcement. (cont'd)						
	b. Cement a 1-3/8 to 1-5/8 inch diameter circular piece of the same coated cloth as the coveralls (see 3.3.1) to the inside and at each end of the collar where the snap fasteners are to be placed. The circular patch shall be centered over the punch hole and placed properly so that it will not interfere with the folding and stitching of the collar.						
*	c. Fold grown-on facing as indicated by marks on pattern. Cement back to back of grown-on facing to within 3/8 to 1/2 inch from raw edge. Place gusset under raw edge, overlapping the edge of the gusset by 3/8 to 1/2 inch. Cement face and back of gusset edge in place. Seam with one row of stitching thru all plies, starting at neck opening on one side down to the bottom of facing. Perform the same operations to opposite side.	301 or 401	SSbc-1	7-9	B	B	
	d. Turn in raw edges of top and bottom of facings 3/16 to 1/4 inch and stitch with one row of stitches across tops and bottoms 1/8 to 3/16 inch from edge.	301 or 401	SSc-1	7-9	B	B	
*	e. Overlap raw edge of body with gusset in the crotch area, approximately 3/8 inch. Cement edges together. Stitch from bottom of facing on one side, across crotch bottom of gusset to bottom of opposite facing.	301 or 401	LSa-1	7-9	B	B	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Attach gusset, facing, and reinforcement. (cont'd)						
	f. Strap inside and outside seams of gusset with 1 inch wide strapping.						
	g. Strap outside of snap cap and snap stud facing seams with 1 inch wide bias cut strapping.						
18.	Attach collar and neck drawcord loop.						
	a. Sew underside of collar to coverall at gorge notching, catching folded neck loop (centered on the back), and corners of gusset in the row of stitching.	301 or 401	LSa-1	7-9	B or E	B or E	
	b. Fold collar in half as indicated by notches on pattern. Turn in edges of collar and stitch 3/16 to 1/4 inch from edge.	301 or 401	SSc-1	7-9	B or E	B or E	
	c. Seam top side of collar to neck.	301	LSa-1	7-9	B or E	B or E	
*	d. Bartack vertically (through all plies) corners of collar facing junction.	Bartack	-	28 per Bartack	B or E	B or E	
	e. Strap all collar seams on the outside only with 1 inch bias strapping.						
19.	Make and attach neck adjustment straps.						
	a. Thread 3/4 inch webbing over center bar of buckle, turn under the raw edge of buckle strap and boxstitch to collar as indicated on pattern (strap to finish 1-1/2 ± 1/4 inches long).	301	LSa-1	7-9	E	E	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Make and attach neck adjustment straps. (cont'd)						
	b. Thread adjusting strap (3/4 inch webbing) over bar next to tongue of buckle, turn in the raw end of strap forming a 3/4 ± 1/16 inch hem and box stitch. Turn in and stitch end of strap to collar, as indicated on pattern with a 5/8 inch box stitch (strap to finish 4-1/2 ± 1/4 inches).	301	LSd-1	7-9	E	E	
	c. Cement 1-1/4 inches circular reinforcement patches of the coated fabric on the inside of the collar to cover all box stitching of buckle strap.						
20.	Attach front snap fasteners.						
	Attach five snap caps on the left front facing and one on collar as indicated by marks on the patterns. Attach five snap studs on the right front facing and one on the collar as indicated by marks on the pattern.						
21.	Attach label.						
	a. Stitch around all four edges of label to a 3-1/4 by 2 inch piece of self material.	301	-	8-10	B or E	B or E	
	b. Cement label assembly to the inside back of the coverall on either the right or left hand side and from 4 to 7 inches below the collar.						
	c. Alternatively the label may be stamped directly on the fabric in the same position referenced above.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Complete assembly. a. Attach suspenders to the inside of the coverall. b. Insert belt through belt loops. c. Insert draw cords through eyelets and drawcord tunnel. d. Attach slide keeper to drawcord.						
23.	Preparation for packaging. A light application of dusting powder shall be applied to the finished coverall (see 3.3.18).						

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3.7 Finished measurements. The finished measurements for the coverall shall conform to table III.

TABLE III. Finished measurements

	Small	Medium	Large	Extra Large	Tolerance
	Inches	Inches	Inches	Inches	Inches
1/2 chest 1/	26	28	30	32	± 3/4
1/2 waist 2/	27-1/2	29-1/2	31-1/2	33-1/2	± 3/4
Sleeve inseam 3/	21	22	23	24	± 3/4
Trouser inseam 4/	31	32	33	34	± 3/4

1/ With the front of coverall snapped and smoothed flat, measure across front of coat part from a point 1/2 inch below armhole seam from side fold to side fold.

2/ With the front of coverall snapped and smoothed flat, measure across waist from side to side fold.

3/ Measure the sleeve inseam from the base of armhole along the inseam of sleeve to bottom of sleeve.

4/ Measure from the base of crotch along the inside of leg to the bottom of leg.

3.8 Performance.

3.8.1 Adhesion of strapping. After application of the strapping as specified in 3.5.3, the adhesion property shall meet the requirements of table IV when tested as specified in 4.4.6.

TABLE IV. Adhesion of strapping

Characteristic	Minimum requirement
Adhesion, pounds per inch of width:	
Initial	3.0
After water immersion	2.0
After heat aging	2.0

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3.9 Leakage. The coverall shall be free from leaks when tested as specified in 4.4.7. After testing, the coveralls shall be cleaned to remove stains resulting from application of the soap solution.

* 3.10 Workmanship. The finished coverall shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable quality levels. Utmost care shall be taken during component and sub-assembly fabrication to ensure quality workmanship and safety of the service person using the item. All cemented areas shall be fully cured prior to packaging.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.4 and 4.4.5 and tested for the characteristics specified in 4.4.6. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on components listed in table V for the characteristics noted. All tests are applicable to the sample unit. The sample unit for testing of preprinted labels shall be two labels; for ink shall be a 1-pint sample from each manufacturer's batch number; for cement shall be a 1-pint sample from each manufacturer's batch number. The batch number for both ink and cement identifies that quantity of material manufactured at the same time using the same raw materials or subjected to some unit chemical or physical mixing process intended to make the final product homogenous. The sample size shall be two sample units. Any test failure shall be cause for rejection of the component lot or batch.

TABLE V. Component tests

Component	Characteristic	Requirement paragraph	Test method
Cement	Cold crack of cement film	3.3.3	4.5.3
	Hydrostatic resistance:		
	Before applying cement	3.3.3	Method 5512 1/ of FED-STD-191
	After applying cement	3.3.3	Method 5512 2/ of FED-STD-191
Cloth label, (when used)	Resistance to fading and tackiness	3.3.17.3	4.5.4

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TABLE V. Component tests (cont'd)

Component	Characteristic	Requirement paragraph	Test method
Ink (when stamped label is used)	Resistance to fading and tackiness	3.3.17.3	4.5.4 3/

1/ Two 8 by 8 inch pieces of the coated cloth specified in 3.3.1 shall be used to determine hydrostatic resistance.

2/ Two 8 by 8 inch pieces of the coated cloth specified in 3.3.1 shall be completely coated on one side with two layers of the cement specified in 3.3.3 and used to determine hydrostatic resistance.

3/ Two labels shall be prepared from each sample unit of ink by stamping the label legend on two scrap pieces of the coated fabric. The pieces need only be large enough to include the legend.

* 4.4.1.2 Component certification. The contractor shall furnish a certificate of compliance for the requirements specified below.

Component	Requirement paragraph
Cement (composition) 1/	3.3.3
Eyelet and washer	3.3.13
Slide keeper	3.3.16
Label	3.3.17.1
Soap	3.3.19
Cleaning agent (solvency)	3.5.3.1

1/ Certification shall be limited to composition. Other characteristics of cement and label shall be tested as specified.

4.4.2 In-process inspection. Inspection of the coveralls shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. An examination shall be made for defects in seams and stitching before application of the seam strapping. This examination shall be made on unstrapped component parts, or unstrapped coveralls as applicable. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.3 Critical defect examination of end items. Prior to performing the end item sampling examination required in 4.4.4, the coverall shall be 100 percent examined for the critical defects in table VI. The 100 percent critical defect determination shall become a part of the contractor's inspection system or quality program. Any coverall found to contain one or more critical defects shall be rejected.

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* 4.4.4 End item visual examination. The end items shall be examined for the defects listed in table VI. The lot size shall be expressed in units of coveralls. The sample unit shall be one coverall. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 15 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot. Inspection shall be accomplished under lighting of 150 foot-candles of illumination at the inspection surface.

TABLE VI. End item visual defects

Examine	Defect	Classification Critical	Major	Minor
Finish:				
Color	Definite variation in color			301
* Cleanliness	Spot, stain, or streak more than 1 inch long which cannot be readily removed with naptha, methanol, or water			302
	Cement line extends more than 1/2 inch on either side of the strapping or major portion of seam.			303
Dusting	Coveralls not dusted, nor not dusted as specified with dusting powder.			304
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein).		201	
Material:				
General	Any component not fabricated of the specified material (unless otherwise indicated herein).		202	
* Coated cloth (including strapping)	Cut, hole, tear which extends through or exposes the base fabric.	101		
	Pit (small spherical depression in coating), abrasion mark or scratch which extends through the coating to a point where base fabric yarns or fabric weave can be seen.	102		
	Uncoated area.	103		

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Material: (cont'd)				
	Blister, tunnel, or delamination of coating.	104		
	Lump or imbedded foreign matter	105		
	Color not uniform			305
	Any pick-off or cavity of any size resembling a pick-off in the coating 1/	106		
	Crease or wrinkle resulting in adhesion of surface or delamination of coating when corrected by manual pressure	107		
	Cracking, crazing, stiff, brittle, dry, burnt, soft, or tacky coating	108		
	Pit which does not expose base fabric		203	
	Any visible abrasion/scratch markings that results in a thinning of butyl/coating but does not expose base fabric		204	
	NOTE: A pick glass may be used to determine the degree of deficiency under "Coated cloth" classifications.			
Construction and workmanship (general, applicable to all components unless otherwise indicated herein):	Any component omitted, damaged, or misplaced, or any operation omitted or not properly performed seriously affecting serviceability		205	
	Any component misplaced, omitted, or damaged, or any operation not properly performed affecting serviceability but not seriously.			306
	Any component damaged or misplaced, or any operation omitted or not properly performed which may result in injury to the user.	109		
	Any component or assembly malformed seriously affecting serviceability.		206	
	Any component or assembly malformed affecting serviceability but not seriously			307
	Any cemented area not fully cured		207	

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Repairs	Patch less than 1-1/4 inches in diameter		208	
	Patch greater than 1-3/4 inches but not greater than 2 inches in diameter.			308
	Patch greater than 2 inches in diameter.		209	
	More than two repairs in either sleeve.			309
	More than eight repairs per coverall.		210	
	More than three repairs on any body part.			310
	Damaged area patched on one side only, except where there is a double layer of the coated cloth.		211	
	Heavily coated side of patch cemented to the coverall.			311
	Patch extends less than 1/2 inch beyond all edges of the damaged area.		212	
	Patches not superimposed by more than 1/4 inch.			312
	Edge lifting of patch, cracking, flaking or removal of the cement or patch when hand flexed as specified in 3.5.6.	110		
	Patch not smoothly and evenly adhered.		213	
	Patch area excessively stiff or tacky affecting serviceability	111		
	Unauthorized repairs.	112		
Seams and stitching	Not seam type specified.		214	
	Not stitch type specified			313
	Stitching omitted where required		215	
	One or two stitches per inch less than the minimum specified			314
	Three or more stitches per inch less than the minimum specified.		216	
	More than the maximum number of stitches per inch specified, but not damaging fabric.			315

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Seams and stitching (cont'd)	More than the maximum number of stitches per inch specified damaging fabric	113		
	One or more rows of stitching forming box stitching omitted or insecure to a degree where assembly may become detached.		217	
	One or more rows of stitching forming box stitching loosely accomplished but assembly is not expected to become detached.			316
	Gage of stitching irregular or not as specified.			317
	Ends of all stitching when not caught in other seams or stitching, backstitched less than 1/4 inch.			318
	Stitching of thread breaks overlapping is less than 1/2 inch at each end of break.			319
	Any raw edges where not permitted.			320
	Any open seam (see note).		218	321
	NOTE: When one or more stitches are broken or two or more consecutive skipped or run-off stitches occur on a single-stitched seam or on both rows of stitching on a double-stitched seam, it shall be classified as a major defect. When one or more stitches are broken or two or more continuous skipped or runoff stitches occur on only one row of a double-stitched seam, it shall be classified as a minor defect.			
Strapped seams	Any open seam completely through coverall.	114		
	Strapping omitted on both sides of any seam where double strapping is required.	115		
	Any seam strapped on only one side where double strapping is required.		219	

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Strapped seams (cont'd)	Strapping on both sides in the same area of seam not securely affixed, i.e., strapping not properly cemented so that it can be readily pulled off seam with little resistance.	116		
	Strapping on only one side of double-strapped seam not securely affixed.		220	
	Strapping not securely affixed on specified single-strapped seams.		221	
	Any blister or delamination in central area of strapping but edges are securely affixed and the bonded portion of strapping is securely cemented to fabric.			322
	Strapping on one side of a double-strapped seam does not completely cover stitching on seam.		222	
	Strapping on both sides in the same area of seam does not cover stitching on seam.	117		
	Strapping off center by more than 1/8 inch but covers seams and stitching.			323
	Any opening at edge of strapping 1/16 inch or more in depth but does not extend to the stitching.			324
	Any opening 1/32 inch or more in width extending to the stitching in the following areas:			
	- any collar seam			325
	- any inner sleeve seam			326
	- from the outer sleeve hem to 3 inches above			327
	- from the leg hem to 3 inches above			328
	- all other seams		223	
	Any strapping less than 15/16 inch wide.			329

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Strapped seams (cont'd)				
	NOTE: The extent of opening (or tunnel) may be determined by insertion of any blunt instrument suitable for this purpose. Penetration of the instrument shall cease immediately if resistance is met.			
	Any individual seam not having a continuous length of strapping.		224	
	Any ends of strapping that meet, not overlapped 1 ± 1/4 inch.		225	
	Any intersection of strapping not overlapped.		226	
	Butting of ends of strapping.		227	
	Piecing of strapping on individual seams.		228	
	Any opening less than 1/32 inch in width extending to the stitching.			330
Metal components	Any metal component (except stud portion of snap fastener or eyelet) missing, reversed on assembly, or not properly affixed to a degree where it may become detached from assembly, or is damaged to a degree that it will not function.		229	
	Any metal component (except stud portion of snap fastener or eyelet) not properly affixed but will adequately be retained on assembly or is damaged but will function as intended.			331
	More than three splits in eyelet or button barrels of snap fastener.			332
	More than one stud part of snap fastener or eyelet missing, damaged, or not sufficiently set to a degree where it may become detached from assembly.		230	
	Any stud part of snap fastener or eyelet missing or damaged.			333

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Metal components (cont'd)	More than one snap fastener or eyelet misplaced but will serve the purpose intended.			334
	Any snap fastener or eyelet misplaced to a degree where it will not serve the purpose intended.		231	
	Any metal component except eyelets not properly clinched resulting in cutting through fabric.	118		
	Eyelets not properly clinched through washer resulting in loose eyelet washer assembly.			335
	Any sharp burr or edge.			336
	Any snap fastener not functioning properly i.e., fails to snap closed, to provide a secure closure, or to open freely.		232	
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
Assembly details				
Collar	Badly twisted, puckered, or pleated at joining of backs or fronts to collar seriously affecting serviceability.		233	
	Twisted, puckered, or pleated at joining of backs or fronts to collar affecting serviceability but not seriously			337
	Collar extension on left end omitted.		234	
Front openings	Facings tight, short, or twisted resulting in excessive puckers or distortion seriously affecting serviceability.		235	
	Facings tight, short, or twisted resulting in puckering or distorted fronts affecting serviceability but not seriously.			338

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Webbing straps, stay pieces, reinforcing patches other than seam strapping, and gusset cord	Any strap omitted.		236	
	End of any strap not turned under and stitched.			339
	Any reinforcing patch or stay piece omitted or not properly attached.			340
	More than one reinforcing patch or stay piece omitted or not properly attached.	119		
	Gusset cord omitted or not of sufficient length, cut ends of cord not dipped or not properly stitched to preclude fraying, or keeper not affixed.			341
	Crotch reinforcing patch missing or not properly attached.		237	
Sleeve tabs and leg adjusting strap	Not assembled as specified.		238	
	Less than the number specified.		239	
	Cap missing from leg adjustment strap		240	
	Assembly not securely cemented or not cemented in right location.		241	
	Elastic piece omitted from leg adjusting strap.		242	
Gusset	Omitted.	120		
	Front badly puckered or pleated at joining to gusset.			342
Sleeve sponge pads	Any omitted, misplaced, or not adequately cemented to coverall to a degree where the pad will become detached in use.		243	
	Any pad not adequately cemented to coverall but is not expected to become detached in use.			343
Neck loop	Omitted.		244	
	Ends not hemmed or too narrow, not permitting cord to function freely through loop.			344

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Marking	Missing, incomplete, incorrect, not legible, not specified type or size, not in proper location, or not accomplished in the specified manner.			345

4.4.5 Dimensional examination. The end item shall be examined for conformance to all dimensions specified in table III. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of coveralls. The sample unit shall be one coverall. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.6 End item testing. The end item shall be tested for the characteristics listed in table VII. All test requirements are applicable to the sample unit. The lot size shall be expressed in units of coveralls. The sample unit shall be one coverall. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0. Any separation of the coating in lieu of strapping adhesion will be considered an invalid test and a new specimen will have to be tested.

TABLE VII. End item tests

Characteristic	Requirement paragraph	Test method	No of determinations per sample unit	Results reported numerically to the nearest
Adhesion of strapping:				
Initial	3.8.1	4.5.2.1	5	0.1 lb.
After water immersion	3.8.1	4.5.2.2	5	0.1 lb.
After heat aging	3.8.1	4.5.2.3	5	0.1 lb.

4.4.7 Final leakage testing. All coveralls which have passed the critical defect examination in 4.4.3 shall be tested for leakage as specified in 4.5.1. Any coverall failing the leakage test shall be rejected. Every coverall which passes the leakage test shall be rinsed to completely remove the soap solution, dried, and talced prior to packaging.

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4.4.8 Packaging inspection. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of container flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Contents	Number of unit packs per shipping container is more or less than required

4.4.9 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

* 4.5.1 Leakage test. The coverall shall be tested from the outside or from the inside by turning the coveralls inside out. The wrist and leg openings of the coverall shall be blocked off from within by tapered plugs of such

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dimensions as to provide a slip fit with neither wrinkle nor undue extension of the fabric. Two of the four plugs shall be provided with nipples, one for attachment to an air line and the other for attachment to a water manometer (inclined or vertical type). As an alternate, padded clamps may be used to seal wrist and leg openings. The clamps must not damage the butyl coating. Two of the clamps shall provide for nipples which will allow for the attachment of an airline and water manometer. The neck closure of the coverall shall be gathered below the drawcord and tied with tape. The coverall shall be air inflated to 35 ± 5.0 mm water pressure. When proper inflation has been achieved, the cover-all shall be completely coated with a 2.5 weight percent arionic surfactant soap solution using distilled water. After a waiting period of at least 30 seconds, all sections of the coverall shall be examined for signs of bubbling. The appearance of any bubbling shall be considered a test failure. If clamps are used for the leakage tests, visually inspect the wrist and leg cuffs to ensure there are no tears or breaks in the butyl coating. Repair any damage in accordance with TM-10-8400-203-23.

4.5.2 Strapping adhesion tests.

* 4.5.2.1 Strapped seams. After the specimens have been aged for 48 hours, the adhesion of strapping to the seams shall be determined as specified in Method 5962 of FED-STD-191 except as follows: The adhesion shall be the average of the three lowest individual load resistances registered in pounds per inch of strapping width on the autographic chart for 3 inches of seam strapping separation in lieu of the average of the five highest peak loads of resistance registered for 3 inches of separation of strapping.

4.5.2.2 Strapped seams after water immersion. Five 6-inch long strapped seam specimen shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at 75° ± 5°F, removed from the water and while still dripping wet, tested for strapping adhesion as specified in 4.5.2.1.

4.5.2.3 Strapped seams after heat aging. Five 6-inch long strapped seam specimens shall be exposed for 7 days in a circulated air oven at 158° ± 2°F, removed from the oven and brought to equilibrium under standard conditions, and then tested for strapping adhesion as specified in 4.5.2.1.

* 4.5.2.4 Test conditions. Results of physical tests obtained under testing conditions defined in FED-STD-191, FED-STD-406 or FED-STD-601 will be acceptable except in dispute cases. In dispute cases, tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in FED-STD-191.

4.5.3 Cold crack test of cement film. An 8 by 8-inch square of the coated fabric specified in 3.3.1 shall be completely coated on one side only with two layers of cement specified in 3.3.3. The cement shall be applied in accordance with the procedure followed in strapping the seams. The test specimen shall be dried and dusted tack free, conditioned for not less than 72 hours at standard

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conditions, conditioned for an additional 4 hours (± 15 minutes) at -20°F, and then tested at -20°F as specified in Method 5874 of FED-STD-191. In the performance of the test, the cement side shall be considered the heavily coated side.

4.5.4 Label resistance to fading and tackiness test. Two labels shall be tested as follows. One label shall be completely submerged in boiling distilled water for 1 hour and then removed and examined. The other label shall be completely submerged in 10-percent (by volume) acetic acid solution at 75° to 80°F for 1 hour and then removed and examined. Any loss of legibility or any signs of fading or tackiness in either label shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

* 5.1.1 Level A preservation. Each coverall shall have metal components cushioned. Cushioning, having a minimum thickness of 1/8 inch, shall conform to type I of PPP-C-1797. One-inch width gummed paper tape shall conform to type III, grade B of PPP-T-45. When tape is applied, it shall not contact the suit. Two by two inch squares of cushioning shall be slit and fitted onto each of the six buttons and secured in place with tape. Neck straps shall be wrapped with cushioning and taped. The metal components of the inner sleeve shall be cushioned. Outer sleeves and legs shall be wrapped with cushioning and taped in such a manner that all metal components are covered on both sides. The gusset shall be drawn closed. A sheet of cushioning having a minimum width of 4 inches shall be placed in the front facing so as to cushion to drawstring keeper and all inside surfaces of snaps. The cushioning shall extend a minimum of 1 inch above the collar and 1 inch below the front facing. The front snaps shall be snapped. The coveralls shall be laid face down and folded to measure approximately 23 inches in length, 18 inches in width, and 2-1/2 inches in depth. One belt and one pair of suspenders shall be placed separately in plastic bags. One folded coverall, one belt and one pair of suspenders shall be unit packed in a snug-fitting, flat style, clear polyethylene film bag conforming to A-A-50083. Each bagged coverall shall be packed in a fiberboard box conforming to style OPF, type CF, variety SW, class domestic, grade 175 of PPP-B-636. Inside dimensions of the box shall be approximately 23 inches in length, 18 inches in width, and 2-1/2 inches in depth. Each box shall have all seams and corners secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Five coveralls of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, five in depth within a shipping container. Inside dimensions of each container shall be approximately 23 1/2 inches in length, 18 1/2 inches in width, and 14

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inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of the container specification except that the inspection shall be in accordance with 4.4.8. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Five coveralls of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, five in depth within a shipping container. Inside dimensions of each container shall be approximately 23 1/2 inches in length, 18 1/2 inches in width, and 14 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification except that the inspection shall be in accordance with 4.4.8.

5.2.2.1 Weather-resistant container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.8.

5.2.3 Commercial packing. Coveralls, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), coveralls packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 90 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coveralls are intended for protection of personnel engaged in extremely hazardous decontamination work or other special operations involving danger from spillage or splashing of liquid chemical agents. This is

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a special purpose Life Support Clothing and Equipment (LSC&E) item. All government administrative and surveillance procedures applicable to LSC&E items shall be invoked in accordance with the contract or purchase order.

* 6.2 Ordering data. Acquisition specifications should specify the following:

a. Title, number, and date of this document.
b. Special provisions for verification inspection of Life Support Clothing and Equipment (see 1.1).
c. Size required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.1, 4.3, and 6.4).
f. Level of packing (see 5.2).
g. Type and class of unit load required (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting offer should include specific instructions in acquisition documents, regarding arrangements for selection, inspection, and approval of the first article.

* 6.5 Soap solution. The anionic surfactant soap "IGEPON-T77" manufactured by GAF Corporation has satisfactorily met the requirements of 3.3.19.

6.6 Subject term (key word) listing.

Chemicals
Decontamination
Overgarment
Protection

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - NU
Air Force - 99

Review activities:

Army - EA, MD
Air Force - 82
DLA - CT

User activities:

Navy - MC, OS, CG
Air Force - 45

Preparing activity:

Army - GL

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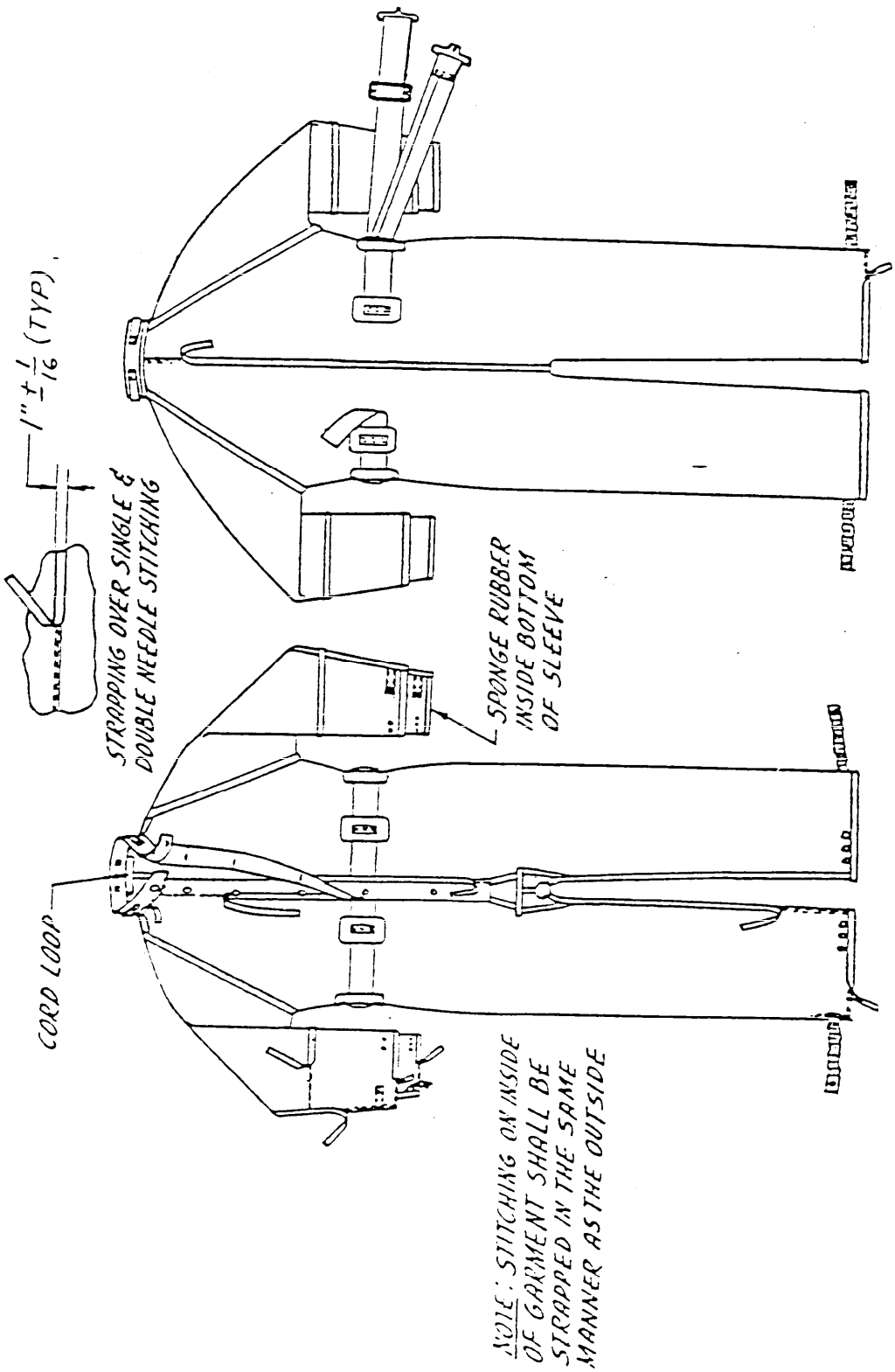


FIG. 1-COVERALLS, TOXICOLOGICAL AGENTS, PROTECTIVE, M-3

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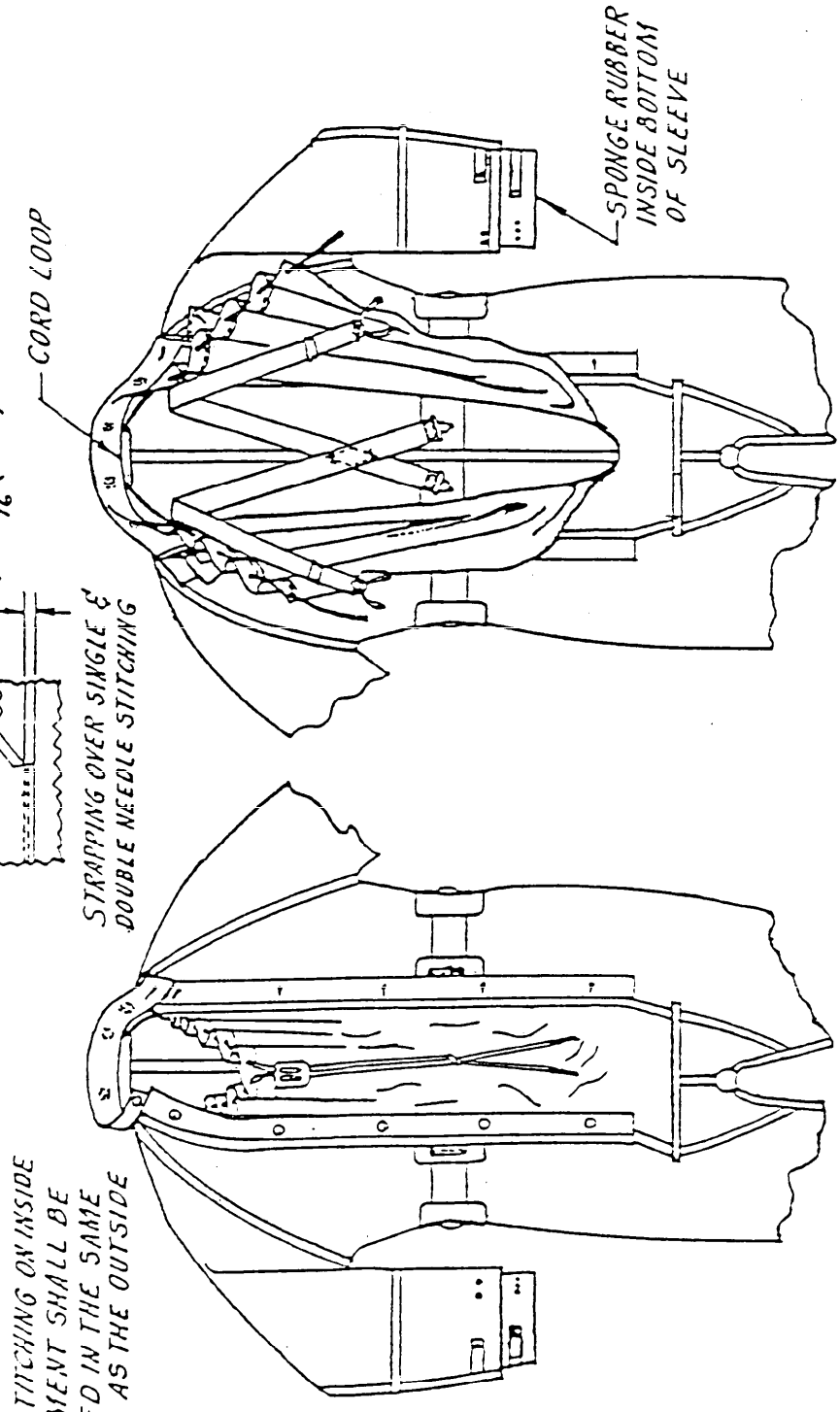


FIG. 2 COVERALLS, TOXICOLOGICAL AGENTS, PROTECTIVE, M-3
2-1-305

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-2181M	1990 September 27

3. DOCUMENT TITLE
COVERALLS, TOXICOLOGICAL AGENTS PROTECTIVE, M-3

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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